

Software Requirements Specification (SRS)

Project: Online Bookstore

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
1.0	11-09-2026	Project Team	Initial SRS draft	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for an Online Bookstore e-commerce platform. It defines functional and non-functional requirements, interfaces, and verification criteria intended for developers, testers, and evaluators to use as a reference.

1.2 Scope

Covers user interactions on the web/mobile storefront for browsing, searching, purchasing, and reviewing books; integration with payment gateways, shipping/logistics providers, and inventory management; and seller and admin operations for catalog and order management. Excludes physical warehouse logistics and third-party courier internal operations beyond API-level integration.

1.3 Audience

Developers, QA Engineers, System Integrators, Project Stakeholders, and Assessment Evaluators.

1.4 Definitions

List of acronyms: SRS, UI, API, JWT, SKU, DB, REST, MERN, ORM, CDN, SSL/TLS, RBAC.

2. Overall description

2.1 Product perspective

The Online Bookstore is a web-based e-commerce platform built on Django (Python), enabling customers to browse, search, and purchase books. At the same time, sellers and admins manage catalog, inventory, and orders through a dedicated dashboard. The platform uses Django's built-in admin interface and ORM to speed up catalog, order, and user management. The platform integrates with external payment gateways, email/SMS notification services, and shipping providers.

2.2 Major product functions (detailed)

- User registration and authentication (buyer, seller, admin)
- Book catalog browsing and search (by title, author, genre, ISBN)
- Product detail view with ratings and reviews
- Shopping cart and wishlist management
- Checkout and payment processing
- Order placement, tracking, and history
- Seller catalog and inventory management
- Admin user, catalog, and order oversight
- Ratings and reviews submission and moderation
- Recommendation of related/similar books
- Email/SMS order notifications
- Discount and coupon code application

2.3 User roles and characteristics (expanded)

- Buyer/Customer: general public with basic web literacy; expects intuitive browsing and a quick, secure checkout.
- Seller: lists and manages own book inventory, prices, and fulfils orders; needs a straightforward dashboard.
- Admin: platform-wide oversight — user management, catalog moderation, dispute resolution, and analytics.

2.4 Operating environment

Web application accessible via modern browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile devices; backend hosted on cloud infrastructure (e.g., AWS/Azure) running Django (Python) with a PostgreSQL database, fronted by a CDN for static assets.

2.5 Constraints

PCI-DSS considerations for payment data handling, applicable data-protection regulations for user data, mandatory HTTPS/TLS for all communications, and rate limits imposed by third-party payment gateway APIs.

3. External interface requirements

3.1 User interfaces

Responsive web UI with clear navigation (home, catalog, cart, account, dashboard). Accessibility features include screen-reader-friendly labels and adequate colour contrast, targeting WCAG 2.1 AA guidance.

3.2 Hardware interfaces

None mandatory. The platform runs on standard client devices (desktop, laptop, or mobile) with an internet connection; no dedicated peripherals are required.

3.3 Software interfaces

- Payment Gateway API (e.g., Razorpay/Stripe) over HTTPS for transaction authorisation and settlement.
- Email/SMS Notification API for registration, order, and shipping updates.
- Shipping/Logistics Partner API for rate calculation and shipment tracking.
- PostgreSQL database accessed through the Django ORM.

3.4 Communications

- TLS 1.2+ enforced for all client-server and server-to-third-party communications.
- Retry/backoff strategy for external API failures, with an asynchronous job queue for notification dispatch.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow BOOK-F-####.

4.1 Authentication & Account Management

Description: Register and authenticate buyers, sellers, and admins, and manage sessions securely.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-001	The system shall allow users to register with an email and password, verified via a confirmation email.	Functional	High	Product/Business	AC-BOOK-F-001: Registration succeeds; account activates only after email verification. Test: TC-Auth-01	Requires Email API
BOOK-F-002	The system shall authenticate users via a hashed password (bcrypt) and issue a JWT session token.	Functional	High	Security	AC-BOOK-F-002: Valid credentials return a JWT; invalid credentials return 401. Test: TC-Auth-02	
BOOK-F-003	The system shall lock an account for 15 minutes	Functional	High	Security	AC-BOOK-F-003: 6th attempt is blocked with a	Audit review required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	after 5 consecutive failed login attempts and log the event.				message; audit log entry created. Test: TC-Auth-03	

4.2 Catalog Browsing & Search

Description: Enable buyers to browse and search the book catalog by multiple criteria.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
BOOK-F-010	The system shall display a paginated catalog of available books with title, author, price, and cover image.	Functional	High	Business	AC-BOOK-F-010: Catalog page loads in ≤2s for 20 items/page. Test: TC-Cat-01	
BOOK-F-011	The system shall allow search by title, author, ISBN, or genre, with filter and sort options (price, rating, newest).	Functional	High	Product	AC-BOOK-F-011: Search returns relevant matches; filters and sort apply correctly. Test: TC-Cat-02	

4.3 Shopping Cart & Wishlist

Requirements: The system shall allow adding and removing items from the cart and wishlist, and persist the cart across sessions for logged-in users. IDs BOOK-F-020..BOOK-F-023.

4.4 Checkout & Payment

Requirements: Support secure checkout with address selection and coupon application, process payment via the integrated gateway, and generate an order confirmation and invoice. IDs BOOK-F-030..BOOK-F-035.

4.5 Order Management & Tracking

Requirements: Provide order history and status tracking (placed, shipped, delivered, cancelled), and return/refund requests for buyers, plus an order-fulfilment dashboard for sellers and admins. IDs BOOK-F-040..BOOK-F-044.

4.6 Seller & Admin Management

Requirements: Provide a seller dashboard for listing and updating books and inventory, and an admin dashboard for user, catalog, and dispute management plus sales analytics. IDs BOOK-F-050..BOOK-F-054.

4.7 Reviews & Ratings

Requirements: Allow verified buyers to submit ratings and reviews on purchased books, display aggregate ratings on product pages, and support moderation by admin. IDs BOOK-F-060..BOOK-F-062.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs BOOK-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
BOOK-NF-001	Catalog and product pages shall load in ≤2 seconds for 90% of requests under normal load.	Performance	High	90th percentile ≤2s in production-like test. Test: TC-Perf-01
BOOK-NF-002	The platform shall provide 99.5% availability monthly; scheduled maintenance windows excluded.	Reliability	High	Uptime reports show ≥99.5% per month. Test: Ops reports
BOOK-NF-003	All payment and cardholder data must comply with PCI-DSS; no card data stored in plaintext on application servers.	Security/Compliance	High	PCI-DSS audit checklist pass. Test: TC-Sec-01
BOOK-NF-004	The system shall support at least 1,000 concurrent users without response times exceeding defined thresholds.	Scalability	Medium	Load test sustains target concurrency within SLA. Test: TC-Perf-02
BOOK-NF-005	The storefront shall conform to WCAG 2.1 AA for key user flows (browse, cart, checkout).	Usability/Accessibility	Medium	Accessibility audit pass. Test: TC-UX-01

5.1. Security

5.1.1 Security Objectives

- Protect user credentials and payment information from unauthorised access and disclosure.
- Ensure the integrity and availability of transaction and order data against tampering or loss.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
BOOK-SR-001	TLS 1.2+ shall be mandatory for all network connections.	Security	High	AC-SR-001: All endpoints reject non-TLS connections. Test: TC-Sec-02
BOOK-SR-002	Passwords shall be stored using an industry-standard salted hash (bcrypt/argon2).	Security	High	AC-SR-002: Password store contains no plaintext values. Test: TC-Sec-03
BOOK-SR-003	Payment card data shall never be stored on application servers; handled via PCI-compliant gateway tokenisation.	Security	High	AC-SR-003: DB review confirms no raw card data present. Test: TC-Sec-04
BOOK-SR-004	Role-based access control (RBAC) shall be enforced for seller and admin functions.	Security	High	AC-SR-004: Unauthorised role access attempts return 403. Test: TC-Sec-05
BOOK-SR-005	All admin actions shall be logged with timestamp, user ID, and action performed, retained for 1 year.	Security	Medium	AC-SR-005: Audit log entries verified for sample admin actions. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: all high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all test cases passed.
- Acceptance test suites: Authentication, Catalog & Search, Cart & Checkout, Order Management, Payments, Security, and Accessibility tests.

7. System models and diagrams

7.1 UML Use-Case diagram

Two use-case diagrams are provided: one for the Buyer role, and one covering the Seller and Admin roles.

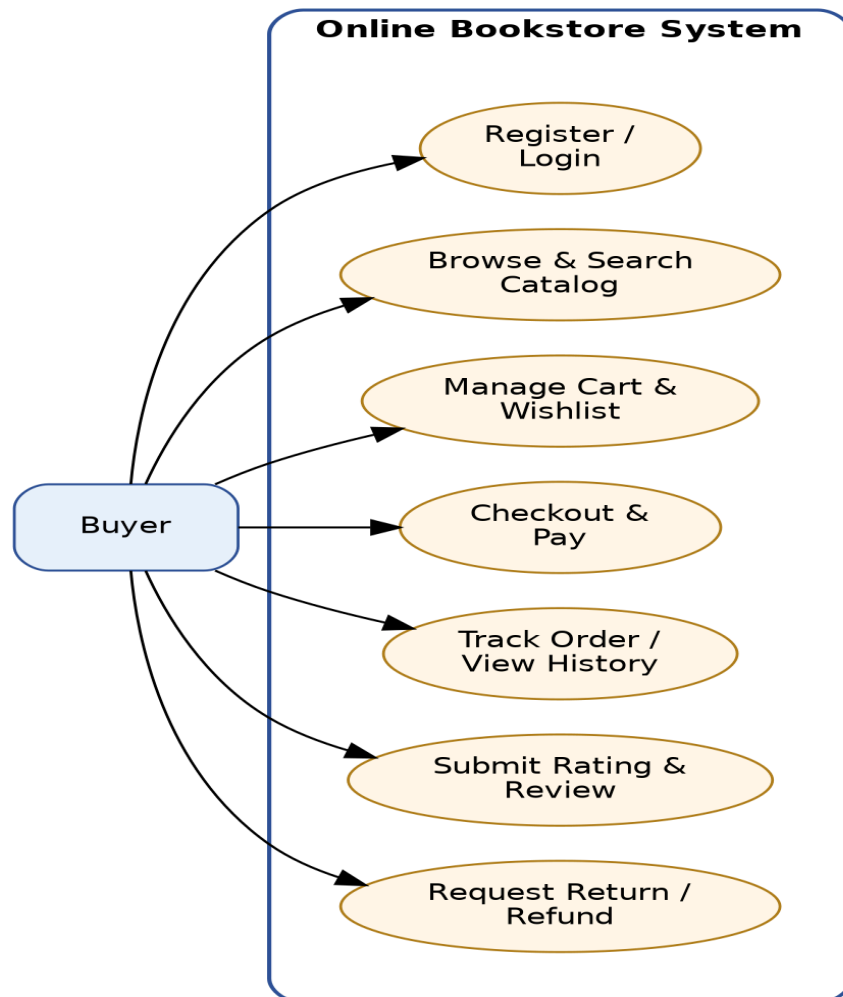


Figure 1: Buyer use-case diagram

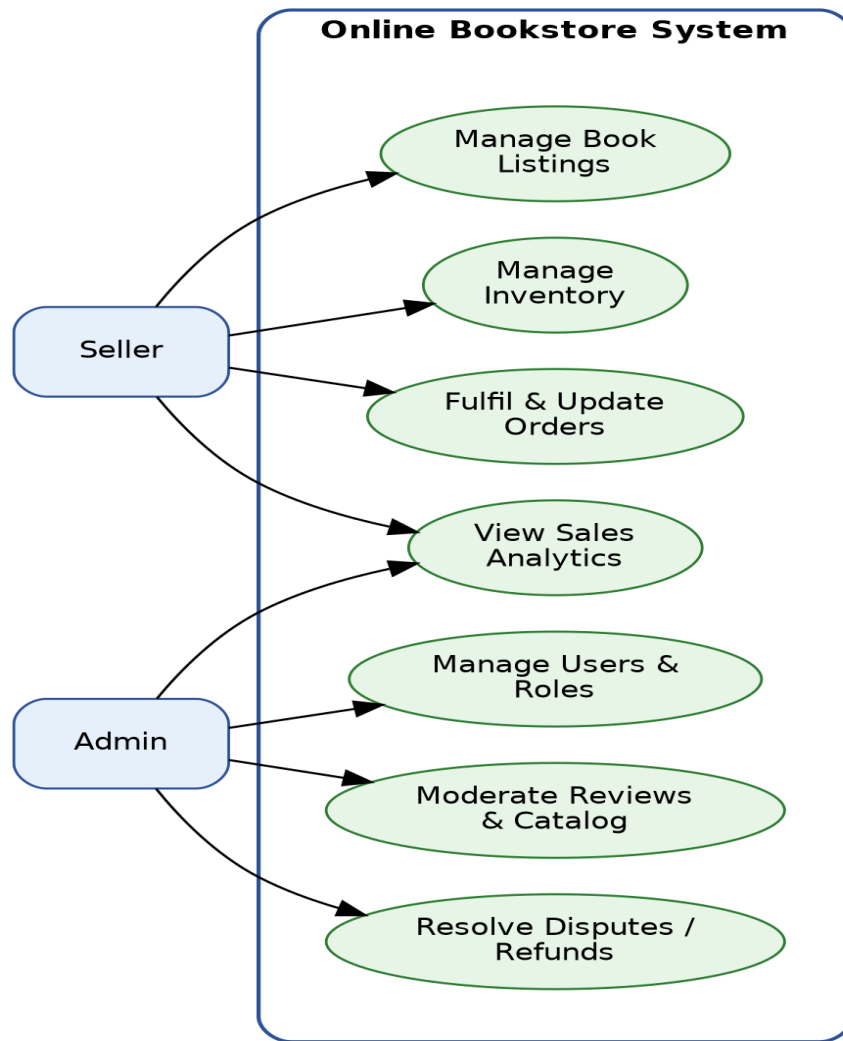


Figure 2: Seller and Admin use-case diagram

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
BOOK-F-001	Register user	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
BOOK-F-002	Authenticate user	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N	
BOOK-F-010	Display catalog	4.2 / DS-Cat-01	CatalogModule	TC-Cat-01	N	
BOOK-F-011	Search & filter	4.2 / DS-Cat-02	CatalogModule	TC-Cat-02	N	
BOOK-N-F-001	Page load target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	
BOOK-S-R-003	No card data stored	5.1.2 / DS-Sec-01	PaymentModule	TC-Sec-04	N	